

Homework (Habanero)

Q1. Simplify $\sqrt{64}$

- (A) 8
- (B) 16
- (C) $\sqrt{16}$
- (D) 4
- (E) $\sqrt{8}$

have doubled in size after 4 hours?

- (A) 8
- (B) 16
- (C) 32
- (D) 4
- (E) 2

Q2. Simplify $\sqrt[3]{27}$

- (A) 3
- (B) 9
- (C) 27
- (D) $\sqrt[3]{9}$
- (E) $\sqrt{3}$

Q6. Simplify $\sqrt{144}$

- (A) 8
- (B) 12
- (C) 16
- (D) 20
- (E) 24

Q3. A patient needs a dosage of $\sqrt{36}$ milligrams of medication. How many milligrams should the patient take?

- (A) 6
- (B) 8
- (C) 12
- (D) 18
- (E) 24

Q7. Convert $x^{3/2}$ into radical form

- (A) $\sqrt[3]{x}$
- (B) $\sqrt{x^3}$
- (C) $x\sqrt{x}$
- (D) $x\sqrt[3]{x}$
- (E) $\sqrt{x^2}$

Q4. Convert $\sqrt[4]{x}$ into rational exponent form

- (A) $x^{1/4}$
- (B) $x^{1/2}$
- (C) $x^{3/4}$
- (D) $x^{2/3}$
- (E) $x^{3/2}$

Q8. A medical researcher is analyzing data on the spread of a disease. The data shows that the number of cases is increasing by a factor of $\sqrt{2}$ per day. How many times will the number of cases have increased after 6 days?

- (A) 8
- (B) 16
- (C) 32
- (D) 64
- (E) 128

Q5. A biologist is studying cell growth. The cells are doubling in size $\sqrt{2}$ times per hour. How many times will the cells

Open Ended Questions (FRQ)

Q9. A patient needs a dosage of $\sqrt[3]{343}$ milligrams of medication. Show all work to simplify the radical and determine the dosage the patient should take.

Q10. Convert the radical expression $\sqrt[4]{x^5}$ into rational exponent form and simplify. Show all work and explain your reasoning.

Chef's Tips

- Tip 1: Remind students to simplify radicals and rational exponents carefully and show all work.
- Tip 2: Encourage students to use real-world examples, such as medicine dosage and cell growth, to make the concepts more relatable and interesting.
- Tip 3: Emphasize the importance of converting between radical and rational exponent form to solve problems in different contexts.